

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI-602105

CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

SUBMITTED BY

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in

COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

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2.1 SUSPECTED UNAUTHORIZED ACCESS TO A COLLEGE SERVER

Scenario: Out-of-hours copying of question-paper files from the CSE department examination server.

Assumptions

A1. The server runs Linux (e.g., Ubuntu) with SSH and a Samba file share. Authentication and file-access logs are enabled.

A2. The server clock is synchronised to an NTP source and set to IST (UTC+5:30). This must be verified, not assumed.

A3. The suspect workstation is a Windows 10 desktop in a staff room. It was found powered on.

A4. A USB flash drive was recovered from the workstation's front port.

A5. Written authorization to investigate has been obtained from the Principal/competent authority before any collection starts.

1. Forensic phases applied to this case

Table 1. Investigation phases and activities

Phase	Activities in this case	Output
Identification	Log the administrator's complaint (who, what, when). Define scope (exam files, time window). Identify evidence sources: server, logs, workstation, USB, network devices, physical-access records. Confirm authorization. Rank sources by volatility.	Evidence source list, scope statement, authorization note
Collection	Document the scene with photographs. Capture volatile data from the running workstation (RAM, network connections, logged-in user) before shutdown. Image the disk and USB behind write blockers. Export server logs as read-only copies. Hash everything and start chain-of-custody forms.	Verified images, log copies, hash list, custody forms
Examination	Work only on forensic copies. Parse file system and log formats. Recover deleted files. Filter irrelevant files (known OS files via hash sets). Search for the exam file names and keywords.	Relevant artifacts extracted from raw data
Analysis	Correlate server logs, workstation artifacts and USB artifacts. Build the timeline. Test alternative explanations. Compare copied files with server originals using hashes.	Timeline, linkage of account to device to files, conclusions
Presentation	Prepare the report (technical and executive versions). Explain the method and limitations. Hand over evidence with custody records. Be ready to explain findings to the authority and, if needed, a court.	Final forensic report, presentation

The phases are iterative. Examination may reveal a new source (e.g., a cloud-sync account), which sends the team back to Identification and Collection.

2. Forensic investigation flowchart

COMPLAINT RECEIVED AND RECORDED

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Confirm written authorization and define the investigation scope

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Identify evidence sources and rank them by volatility

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Secure and photograph the scene; isolate a live workstation from the network where appropriate

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DECISION: Is the workstation powered on?

YES → Capture volatile data (RAM, sessions, connections), then follow a controlled shutdown plan

NO → Keep powered off; document its state and do not boot it

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Acquire forensic images of the disk and USB through write blockers; preserve server logs

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Calculate and verify hashes; document chain of custody

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Examine verified copies; recover, filter, and search relevant artifacts

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Correlate artifacts, build a normalized timeline, and test alternative explanations

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Prepare report and plain-language summary; present findings and hand over evidence

Figure 1. Investigation flow from complaint to presentation

3. Potential digital evidence and investigative value

Table 2. Evidence items and classification

ID	Evidence item	Source	What it can show	Investigative value	Volatility
E1	Server authentication logs (e.g., SSH/system auth logs)	Server	Which account logged in, from which IP, at what time	High (primary): establishes the access event	Medium (rotation risk)
E2	File-share/audit logs of the exam directory	Server	Which files were opened/copied/modified, by which account	High (primary)	Medium
E3	Suspect workstation disk	Workstation	User profile, recent files, executed programs, USB connection traces, deleted files	High (primary)	Low (persistent)
E4	Workstation RAM and network connections	Workstation (live)	Active sessions, running processes, remote connections	High but fragile	Very high

ID	Evidence item	Source	What it can show	Investigative value	Volatility
E5	Recovered USB device	USB	Copied exam files, file-system timestamps, device serial	High (primary): possible exfiltration medium	Low
E6	Copied exam files vs. server originals	USB / workstation	Hash match proves the copies came from the originals	Corroborative	Low
E7	Network/firewall/DHCP logs	Network devices	Links an IP address to a device (MAC) at a given time	Corroborative	Medium
E8	Physical access records (door log, CCTV, attendance)	Admin office	Presence of persons in the room at the time	Contextual / supporting	Medium (overwrite)
E9	Administrator's complaint	Complainant	Starting point, reported time window	Contextual (lead)	Low

Classification key:

- **Primary** evidence directly records the event.
- **Corroborative** evidence confirms a primary item from an independent source.
- **Contextual** evidence gives background or leads but cannot prove the act alone.

Volatility ordering follows RFC 3227: RAM and network state first, then logs and temporary files, then disk contents.

4. Precautions during collection and preservation

Table 3. Precautions and their purpose

Precaution	Why it matters
Obtain written authorization first	Keeps collection lawful and defensible
Photograph and sketch the scene, screen state, cables and USB ports	Proves the original state
Collect in order of volatility (RAM, network state, logs, disks)	Prevents loss of fragile data
Do not browse the suspect workstation or open files on it	Avoids changing timestamps and artifacts
Isolate from the network (disable the network port) rather than shutting down first if the system is live	Stops remote tampering while preserving RAM
Use hardware write blockers for the internal disk and USB	Prevents modification of the original media
Create a bit-for-bit forensic image; verify with hash (e.g., SHA-256) before and after	Demonstrates integrity
Export server logs by read-only copy, hash them, and record log-rotation settings	Prevents alteration and shows completeness
Record the clock offsets (server, workstation, network devices) versus a trusted reference	Needed for a reliable timeline
Label, bag (anti-static for drives), seal and store evidence securely	Prevents tampering and damage
Maintain a continuous chain-of-custody log	Shows who held the evidence at each stage
Perform all analysis on verified copies; keep originals sealed	Original is the master record
Maintain contemporaneous notes of every action with date/time and examiner name	Allows the work to be reproduced and defended

5. Establishing the timeline of unauthorized access

Method

1. List every timestamp source: server auth log, share audit log, workstation file-system timestamps (created/modified/accessed), registry/event artifacts (USB first/last connected), USB file timestamps, network logs, door/CCTV logs.
2. Record each source's clock offset and time zone, and convert all entries to a single reference (e.g., UTC, with IST shown alongside).
3. Merge entries into one sorted table with columns: time, source, event, account/device, confidence.
4. Identify the *access window* from the server logs (login, file access, logout).
5. Look for the matching events on the workstation (remote session, file copy, USB insertion) inside that window.
6. Corroborate with independent sources (network logs, physical access records).
7. Mark gaps and conflicts rather than hiding them.

Table 4. Sample timeline structure

Time (IST)	Source	Event	Account/Device	Role in analysis
21:58	Door/attendance log	Room entry recorded	Card ID *X*	Contextual
22:03	Network/DHCP log	IP *a.b.c.d* leased to workstation MAC	Workstation	Links IP to device
22:07	Server auth log	Successful login from *a.b.c.d*	User *U*	Primary
22:10–22:14	Share audit log	Exam directory files opened/read	User *U*	Primary
22:12	Workstation registry/event artifact	USB device connected (serial *S*)	Workstation	Corroborative
22:15	USB file-system data	Files with exam names created	USB	Corroborative
22:19	Server auth log	Session logout	User *U*	Primary

Timestamps from different systems need alignment. A small clock offset can change the order of events.

6. Final forensic report and presentation to a non-technical authority

Sample forensic report structure

1. Cover page: case ID, title, examiner, date, classification (confidential)
2. Executive summary (half page, plain language)
3. Authorization and scope
4. Background and complaint
5. Evidence inventory (item, ID, description, hash, source)
6. Methodology and tools (versions, write blockers, hashing algorithm)
7. Chain-of-custody summary
8. Findings (each finding: what, where, evidence ID, confidence)

9. Timeline of events
10. Analysis and alternative explanations considered
11. Limitations and assumptions
12. Conclusion (facts supported by evidence)
13. Recommendations (e.g., log retention, access control, USB policy)
14. Appendices: hash lists, custody forms, tool outputs, glossary, examiner declaration

Presenting to a non-technical authority

- Open with a one-paragraph answer: what happened, how sure we are, and what is not known.
- Use plain language and everyday analogies (e.g., a login record is like a visitor register).
- Show one visual timeline rather than raw logs.
- Separate **facts** (supported by evidence), **interpretation**, and **opinion**.
- State confidence levels and limitations honestly, avoiding statements like "the user is guilty". The report states what the evidence shows.
- Offer recommendations the authority can act on, such as policy changes.
- Keep technical detail in appendices for those who want it.

7. Legal and ethical considerations

Table 5. Legal and ethical considerations (college environment)

Area	Consideration
Authority	Act only with written authorization from a competent authority; institutional IT/acceptable-use policy should permit inspection of institutional systems
Legal framework (India)	Unauthorized access and copying may fall under the Information Technology Act, 2000 (e.g., Sections 43 and 66). Admissibility of electronic records is governed by Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, which replaced the Indian Evidence Act, 1872 from 1 July 2024. The Section 63 certificate must disclose the hash value of the electronic record, so hashing at collection is a legal requirement in practice, not only good practice
Privacy and data protection	The workstation and server may hold students' personal data. Collect and examine only what is relevant, and follow the Digital Personal Data Protection Act, 2023
Proportionality	Do not examine personal files unrelated to the complaint
Confidentiality	Share findings on a need-to-know basis; protect exam material and personal data
Impartiality	Examiner must remain neutral, record exculpatory as well as incriminating evidence, and avoid presuming guilt
Chain of custody and integrity	Required for evidence to be accepted and trusted
Escalation	If criminal conduct is likely, involve law enforcement and legal counsel early

Conclusion

A defensible investigation of the server incident depends on authorized, ordered collection, verified imaging, correlation of server and workstation evidence on a normalised timeline, and a report that separates facts from interpretation. Hashing and the chain of custody make the evidence reliable and defensible. Any attribution to a person must acknowledge limits such as shared credentials and clock error.

References

1. Kent, K., Chevalier, S., Grance, T., Dang, H. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
2. Brezinski, D., Killalea, T. (2002). *Guidelines for Evidence Collection and Archiving*, RFC 3227, IETF.
3. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
4. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
5. Information Technology Act, 2000 (India); Bharatiya Sakshya Adhiniyam, 2023, Section 63; Digital Personal Data Protection Act, 2023.

2.2 LINKING A SUSPICIOUS FILE TO A USER USING FILE-SYSTEM EVIDENCE

Selected scenario: Attributing a disguised executable and an encrypted archive to one of several staff accounts on a shared lab desktop.

Assumptions

A1. The desktop runs Windows 10 on an NTFS-formatted internal drive with three standard local accounts (**staffA, staffB, staffC**) and a local administrator account. All names are **hypothetical**.

A2. A forensic image has already been acquired with write protection and verified by hash.

A3. The suspicious items are an executable with a document-style name, several files with the hidden attribute, an encrypted archive, and browser profile data.

A4. ILookIX is the analysis tool. Menu names vary by version, so the workflow below describes functions, not exact screens.

1. Hardware and software components to document before analysis

Table 1. System inventory table

Category	Component to document	Why it matters	Example entry (Hypothetical placeholder)
Hardware	Make, model, serial number, asset tag	Identifies the exact device	*Brand/Model, Serial 1234*
Hardware	CPU, RAM size	Context for memory capture and performance	*Specs*
Hardware	Storage devices: type (HDD/SSD), capacity, interface, model, serial	Identifies source media; SSD behaviour (TRIM) affects deleted-data recovery	*1 x SSD, 256 GB*
Hardware	Partition scheme (MBR/GPT), partitions, unallocated/hidden areas	Hidden partitions can hold evidence	*GPT, 3 partitions*
Hardware	External ports and attached/previously attached devices	USB and external storage leads	*4 x USB*
Hardware	Network adapter(s), MAC addresses	Links to network logs	*MAC (recorded)*
Hardware	BIOS/UEFI date/time versus reference time	Clock accuracy for timestamps	*Offset recorded*
Software	OS name, version, build, install date	Determines artifact locations and formats	*Windows 10, build recorded*
Software	Time zone and regional settings	Needed to interpret timestamps	*IST*
Software	File system(s)	NTFS provides $MFT, timestamps and ADS	*NTFS*
Software	User accounts (names, SIDs, last logon)	Core of attribution	*staffA, staffB, staffC*
Software	Installed applications, browsers, encryption/archiving tools	Explains the executable and encrypted data	*List from system*
Software	Security settings: BitLocker/EFS, antivirus, audit policy	Affects accessibility and logging	*Status recorded*

Category	Component to document	Why it matters	Example entry (Hypothetical placeholder)
Forensic tooling	Write blocker model, imaging tool and ILookIX versions, hashing algorithm	Reproducibility and defensibility	*Versions recorded*

2. How storage devices, operating systems, file systems and metadata help locate evidence

Table 2. Layers that guide the search

Layer	What it contains	How it helps locate evidence
Storage device	Physical sectors, partitions, unallocated space	Shows what exists beyond visible files: hidden partitions, deleted data. SSD TRIM may limit recovery
Operating system	Per-user profile folders, registry hives (e.g., NTUSER.DAT), event logs, Prefetch, recent-file and shortcut (LNK) data, Jump Lists	Records user and system activity, giving the "who used what" context
File system (NTFS)	Master File Table ($MFT) with timestamps, file owner/permissions, alternate data streams, journals ($UsnJrnl, $LogFile)	Shows file existence, paths, creation/modification/access/MFT-change times, and often deleted-file remnants
File metadata	Document author/last-saved-by, application-specific data, executable header details, image EXIF	Links content to a creator, tool, time or device
Application data	Browser history, downloads, cache, cookies	Shows where a file came from and how it was used

Practical use: start with the user-profile paths and timestamps to narrow the search, then use metadata and registry artifacts to confirm who interacted with a file.

3. Hidden and encrypted files: effect on the investigation and handling

Table 3. Hidden and encrypted content

Item	Effect on investigation	Appropriate handling
Files with hidden/system attribute	Not shown in normal views, so easily missed. The attribute is only a flag, not proof of concealment intent	Include hidden items in all listings. Record the attribute without drawing intent conclusions
Renamed extensions (e.g., executable named like a document)	Hide true type from casual inspection	Run file-signature (header) analysis to detect mismatches
Alternate data streams and slack/unallocated space	Can hold data not visible in the directory listing	Examine through file-system-aware tools. Carve if needed
Encrypted containers/archives/documents	Content inaccessible. Keyword search and hash-set filtering fail. Metadata (name, size, timestamps, path) is still available	Preserve and hash the encrypted file. Do not alter it. Document its properties. Look for lawful ways to obtain keys (e.g., stored credentials, key material in memory capture, or legal disclosure)
Whole-volume or user-level encryption (e.g., BitLocker, EFS)	May make the image unreadable without keys	Check for encryption before powering off a live system. Capture RAM where possible
Password cracking or compelled disclosure	May be time-consuming and raises legal issues	Do so only under proper authorization. Take legal advice on compelled disclosure

General handling points: record everything, preserve originals, work on copies, do not conclude intent from a hidden or encrypted state alone, and separate "present" from "used by whom".

4. Evidence-linkage matrix

Table 4. Evidence-linkage matrix: correlating artifacts to a user (Hypothetical examples; actual findings depend on the real image)

#	Artifact	Where found	What it records	Link to user	Strength
L1	File path of the suspicious executable	File-system listing	Location, e.g., inside a specific user's profile folder (*C:\Users\staffB\...*)	Placement in a profile folder suggests that account	Moderate
L2	NTFS owner / security descriptor	File properties / $MFT	Owner SID of the file	Maps to an account via SID	Moderate: shows the account that created it, not the person
L3	Created/modified/accessed timestamps	$MFT	When the file appeared and changed	Compared with that account's logon sessions	Supporting
L4	Registry execution history (e.g., UserAssist, RunMRU) in that user's NTUSER.DAT	User registry hive	Programs run by that account, run count, last run	Per-account evidence of execution	Strong
L5	Prefetch entry for the executable	Windows Prefetch	Executable ran on the system, and when	Machine-wide: does not name the user	Supporting
L6	Recent items/LNK/Jump List for the archive or files	User profile	Files opened by that account	Per-account interaction	Strong
L7	Security event log logon/logoff events	Windows event logs	Which account was logged on and when	Places an account at the console in the time window	Strong when timed with L3–L6
L8	Browser history/downloads/cache	That user's browser profile	Sites visited, files downloaded (source URL, time, save path)	Origin of the file, tied to the profile	Strong for download evidence
L9	Document metadata (author, last saved by)	Inside document files	Name or username embedded by the application	Can name a creator	Moderate. Names can be edited or copied
L10	Encrypted archive properties	File system	Name, size, timestamps, location	Shows existence and location; content unknown	Supporting

Correlation principle: one artifact rarely proves attribution. Confidence grows when independent artifacts agree on account, time and path.

5. Workflow for importing/analyzing evidence in ILookIX

Figure 1. ILookIX analysis workflow

CASE SETUP: Record case ID, examiner, authorization, and scope

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EVIDENCE IMPORT: Acquire using an approved workflow or import a verified forensic image/ASB container

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INTEGRITY CHECK: Compare the image hash with the acquisition record

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PROCESSING: Parse file systems, list active/deleted files, hash files, apply known-file filtering, check signatures, index keywords, and extract relevant artifacts

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TRIAGE AND REVIEW: Filter by user profile, hidden/system attributes, signature mismatches, and encrypted-file indicators

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ARTIFACT ANALYSIS: Review registry artifacts, event logs, LNK/Jump Lists, browser data, and other relevant records

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CORRELATION: Compare timestamps and account activity; normalize time zones and document uncertainty

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FINDINGS AND REPORT: Bookmark findings with notes; report hashes, tool/version details, and limitations

signature mismatch, encrypted-file indicators

Step notes

- **Step 1:** the source media is never touched. All analysis runs on the image.

- **Step 2:** the matching hash demonstrates the image was not altered after acquisition.
- **Step 3:** hash-set filtering reduces noise from standard OS files. Signature analysis exposes the executable disguised as a document.
- **Step 4:** encrypted files are identified by signature or properties and are preserved without modification.
- **Step 5:** each bookmark records what it is, where it was found and why it matters, supporting reproducibility.
- Note the ILookIX version and the exact functions used in the report. Do not claim results that were not generated.

6. Direct linking artifact vs. supporting/contextual artifact

Table 5. Direct vs. supporting artifacts

Aspect	Directly linking artifact	Supporting/contextual artifact
Definition	Ties a specific account to a specific action (create, run, download, open)	Shows that something existed or was possible, without tying an account to an action
Examples (this case)	Execution record for the executable in *staffB*'s own registry hive. Browser download record in *staffB*'s profile for the file. Recent-items entry in *staffB*'s profile for the archive	File simply residing in a shared folder. File timestamps alone. Prefetch entry (machine-wide). Hidden attribute. Installed encryption software. Machine logon log without activity detail
Strength	Strong, but still account-level	Weak alone, useful for context and corroboration
Use in report	Primary finding	Background or corroboration

Even a direct artifact links an **account** to an activity, not necessarily a **person**. Linking the account to a person needs additional evidence such as physical presence or authentication records.

7. Limitations and alternative explanations

Table 6. Limitations to consider before attribution

#	Limitation / alternative explanation	Why it matters	How to address it
1	Shared or weak credentials; the account owner may not be the actual user	An account is not a person	Seek corroboration: access logs, physical presence, other devices
2	Malware or remote access could have created or run the files	Activity could occur without the user's knowledge	Examine for malicious or remote-access software, and external connections
3	Timestamps can be changed (e.g., timestamp manipulation), copied, or altered by the clock/time zone	Weakens timeline conclusions	Compare the $STANDARD_INFORMATION and $FILE_NAME times, journals and event logs; document clock offsets
4	Files can be copied from another machine or USB, preserving original times	Creation time may not mean creation on this device	Check for USB artifacts and download records
5	Administrator or another user could have placed files in someone else's profile	Path alone does not prove ownership	Use execution and logon artifacts together with path
6	Encrypted content cannot be read	Purpose and content remain unknown	State this in the report. Avoid assumptions about content
7	Deleted data may be overwritten and tool interpretation can err	Incomplete picture	Validate with a second tool or method; record tool versions

Conclusion

The suspicious executable and encrypted archive can be linked to a user account by correlating file paths, NTFS owner information, timestamps, registry execution traces, event-log logons and browser records. The strongest case comes from several independent, per-account artifacts that agree in time. Limitations such as shared credentials, malware, copied timestamps and unreadable encrypted content must be stated, so the report attributes the activity to an *account* with a stated level of confidence, not to a named *person* without further evidence.

References

1. Carrier, B. (2005). *File System Forensic Analysis*. Addison-Wesley.
2. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
3. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
4. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
5. ILookIX / IXImager documentation (Perlustro LP): consult the version installed in your lab and cite its exact version.

2.3 FORENSIC IMAGING AND SAFEKEEPING OF A SUSPECT STORAGE DEVICE

Selected scenario: Imaging a seized laptop and an attached USB drive in a suspected research-data theft.

Assumptions

A1. A laptop with one internal SATA SSD/HDD and one attached USB flash drive is seized from a staff desk. Both are covered by a written seizure authorization.

A2. The laptop was found powered off, lid closed, with the USB drive plugged in. If it were on, RAM capture would come first (see Table 1, step 5).

A3. Hardware write blockers, a sanitised destination drive, and IXImager (ILookIX suite) are available in the lab.

A4. IXImager menu names vary by version, so the workflow describes functions, not exact screens. The ASB container is described by purpose; confirm its exact fields in your installed version.

A5. All identifiers (serials, case numbers, names, dates) in tables are **hypothetical placeholders**.

1. Procedure to collect and transport the laptop and USB device

Table 1. Collection and transport procedure

Step	Action	Purpose
1	Confirm authorization; open a case number; assign an evidence collector	Lawful, documented start
2	Photograph the scene and the devices in place (screen, ports, cables, USB position, labels)	Records original state
3	Note state: powered on/off, lid, power source, visible connections	Decides live collection vs. shutdown handling
4	Do not power on, log in, or open files	Avoids changing timestamps, logs and artifacts
5	If found powered on: capture RAM and note running state, check for disk encryption (e.g., BitLocker), then follow a controlled shutdown plan	Volatile data and encryption keys disappear at power-off
6	Remove the USB drive only after photographing it in place; do not open it	Preserves the original attachment state
7	Record make, model, serial number and asset tag of the laptop, internal drive (if accessible) and USB drive	Unique identification
8	Disconnect power as needed and prevent wireless connections (e.g., Faraday bag); only trained staff should remove internal network cards	Prevents remote wiping or modification
9	Place each item in a separate anti-static bag; seal with tamper-evident tape; label with case number, item ID, date/time, collector's name and signature	Prevents mix-ups and shows tampering
10	Pack with padding in a rigid container; keep away from magnets, heat and moisture	Prevents physical damage and data loss
11	Start the chain-of-custody form at the seizure point; every hand-over is signed	Continuous accountability
12	Transport directly to the lab; do not leave the items unattended; log departure and arrival times	Prevents unauthorized access

Step	Action	Purpose
13	On arrival: check seals against the form, photograph, log condition, store in a locked evidence room	Confirms integrity at receipt

2. Physical acquisition and why analysis uses an image

Physical acquisition is a sector-by-sector (bit-stream) copy of the entire storage device, including the partition table, unallocated space, file-system slack space, deleted-file remnants and hidden areas. A logical copy captures only the files the operating system presents, so it misses much of this.

Why analyse the image and not the original:

- **Integrity:** the original stays untouched. Even booting the system or mounting the drive normally changes timestamps and logs.
- **Repeatability:** any examiner can re-run the analysis on a verified copy and get the same results.
- **Risk:** a drive may fail, so analysing the original risks irreversible loss. A verified image can be copied again.
- **Defensibility:** courts and reviewers expect the original to remain unaltered, with the image proven identical by hash.
- **Safety:** analysis tools may crash or write temporary data. This does no harm to a copy.

Caveats: physical acquisition of SSDs may not capture data already erased by TRIM or garbage collection. If the drive is encrypted, the image is as unreadable as the original unless keys are available.

3. Imaging workflow using IXImager

Figure 1. Acquisition flowchart

AUTHORIZATION: Open the case and confirm authority to seize and examine the devices

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IDENTIFY SOURCE: Record device make, model, serial number, capacity, and condition

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PREPARE DESTINATION: Use sanitized destination media with sufficient capacity

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PROTECT SOURCE: Connect through a tested hardware write blocker and confirm read-only status

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CONFIGURE IXIMAGER: Record case details, tool version, time references, output format, and hash settings

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ACQUIRE IMAGE: Image the physical device and record progress, read errors, and bad sectors

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VERIFY INTEGRITY: Compare source and image hashes; investigate any mismatch and repeat acquisition if appropriate

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PRESERVE: Complete logs, create a verified working copy, seal originals, and update chain of custody

Table 2. Imaging procedure (IXImager)

Stage	Detailed steps	Record in the acquisition log
Source identification	Read the device label and compare with the seizure record. Connect through the write blocker and note how the device is reported (model, serial, capacity, sector count)	Source description, serial, capacity
Write protection	Use a hardware write blocker (preferred). Check indicators and confirm the source shows as read-only before imaging. For USB, use a USB write blocker, not only an operating-system setting	Blocker make/model/firmware, test result
Destination preparation	Use a forensically sanitised drive with enough free space. Format it and record the file system	Destination serial, sanitisation method
Tool setup	Start IXImager. Select the physical device as the source (not a logical volume). Enter case number, evidence number, examiner and description. Choose the ASB container format and options such as compression and verification, where offered	Tool version, settings
Imaging	Start acquisition. Monitor progress. Note any read errors or bad sectors reported and how the tool handles them	Start/end time, errors, sectors affected
Verification	Let the tool verify the image or recompute its hash. Compare against the hash of the source calculated during acquisition	Hash algorithm, source hash, image hash, match result
Hashing	Use a strong algorithm (SHA-256, optionally with MD5 for compatibility). Record in the log and on the custody form	Hash values (record real values only from the tool)
Documentation	Complete the acquisition log, photograph the setup, sign and date. Make a working copy for analysis, and keep the master sealed	Examiner, date/time, time zone, notes

Repeat the same procedure for the USB drive, as a separate evidence item with its own ID.

4. Hash values and integrity before and after acquisition

A hash function converts data of any size into a fixed-length value. Any change to the input, even one bit, gives a completely different output, so a hash acts as a digital fingerprint of the data.

Table 3. Use of hash values through the evidence lifecycle

Point	Action	What a match shows
Before/during acquisition	Calculate the hash of the source device (through the write blocker)	Baseline fingerprint of the original
Immediately after acquisition	Hash the image and compare with the source hash	Image is a bit-exact copy
On transfer or copy	Re-hash the image after copying to a new drive or sending it	Copy was not corrupted or altered in transit
Before analysis	Verify the image hash again at the start of examination	Evidence is unchanged since acquisition
Before reporting/court	Verify again and quote the hash in the report	Integrity is demonstrated at every stage

- A mismatch means something changed or failed (corruption, bad sectors, or tampering). It must be investigated and explained, not ignored.

- If the source has unreadable sectors, repeated reads may not give identical hashes. Record this in the log, since it explains why the match may not be perfect.
- MD5 and SHA-1 have known collision weaknesses, so SHA-256 is preferred. Recording two algorithms gives extra assurance.
- In India, the certificate under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 requires the hash value of the electronic record, so hashes must be recorded at acquisition (see Assignment 2(1)).

5. Role of the ASB container

An **ASB container** is the evidence-image format used by the ILookIX/IXImager suite. It packages the acquired data together with descriptive and integrity information, so that evidence can be stored, moved, verified and imported into ILookIX as a single managed item. Confirm exact capabilities (compression, segmentation, metadata fields, any protection options) in the documentation for your installed version.

Table 4. Information that should accompany the evidence

Item	Detail
Case and evidence IDs	Case number, evidence/item number
Source device description	Make, model, serial, capacity, interface, where seized
Examiner and authority	Examiner name, organisation, authorization reference
Acquisition details	Tool name and version, date/time and time zone, write blocker used, settings (format, compression)
Integrity data	Hash algorithm(s) and values for source and image; verification result
Error log	Bad sectors/read errors and how they were handled
Time references	Device clock vs. reference time at acquisition
Chain-of-custody record	Copy of the form and storage location
Copies	Number, location, and hash of each copy

6. Data carving

Data carving recovers files from raw data by searching for known file signatures (headers, and where defined, footers) instead of relying on file-system records. Examples: a JPEG begins with the bytes FF D8 FF and ends with FF D9. A PDF begins with "%PDF" and ends with "%%EOF".

When carving is needed

- The file system is damaged, corrupted or reformatted, so directory entries and the master file table are gone.
- Files were deleted and their metadata overwritten, but content remains in unallocated space.
- The partition table is missing or corrupted.
- Data sits in unallocated space, slack space or a memory/page file.

Limitations

- Carved files usually lose file names, paths and timestamps.

- Fragmented files may be recovered only partly or wrongly joined.
- Carving can produce false positives and unrelated files, so results must be validated (open the file, check the hash, compare context).
- Files without clear signatures are hard to carve.

7. Date and time problems in digital evidence

Table 5. Time issues and how to document them

Problem	Why it matters	How to document
Time zone differences (local time vs. UTC)	Systems store times differently; NTFS keeps UTC, FAT keeps local time	Record each source's time zone and convert to one reference
Incorrect system clock or BIOS time	Timestamps may be shifted	At acquisition, record the device clock beside a trusted reference time (e.g., a checked clock or NTP source) and the offset
Daylight saving changes	One-hour shifts	Note whether DST applied and when
Different timestamp resolution (e.g., FAT 2-second resolution)	Small differences may be artifacts	Note file-system characteristics
Timestamps changed by copy, move, or extraction	Creation time may reflect the copy, not the origin	Interpret together with other artifacts; avoid single-timestamp conclusions
Last-access times not always updated	Absence of an update does not mean no access	State this limitation
Deliberate manipulation	Times can be altered by the user	Compare multiple sources (journals, event logs, logs on other devices)

Documentation practice: write the device time and reference time with the offset in the acquisition log, state the time zone used in every table of the report, and describe uncertainty in plain terms (e.g., "times shown are UTC+5:30 after correcting a clock offset of N minutes noted at acquisition").

8. Chain-of-custody form and table for the seized laptop

Table 6. Chain-of-custody form (header)

Field	Entry (Hypothetical placeholder)
Case number	CASE-1056
Evidence item ID	LT-01 (laptop); HD-01 (internal drive); USB-01 (USB drive)
Description	Laptop, make/model, serial XXXX, colour, condition, seal number
Seized from / location	Staff desk, room/department
Seized by	Collector name, designation
Authorization reference	Written authorization no. and date

Table 7. Chain-of-custody table (Hypothetical sample entries; replace with your own)

#	Date/time	Item	Released by	Received by	Purpose / location	Condition / seal	Signatures
1	D1 10:15	LT-01, USB-01	Room staff / scene	Collector (seizure)	Seizure at scene	Bagged and sealed, seal no. S1	Both
2	D1 11:00	LT-01, USB-01	Collector	Transport officer	Transport to lab	Seals intact	Both

#	Date/time	Item	Released by	Received by	Purpose / location	Condition / seal	Signatures
3	D1 12:10	LT-01, USB-01	Transport officer	Evidence custodian	Receipt at lab, locked store	Seals checked, photographed	Both
4	D2 09:30	LT-01 (internal drive removed), USB-01	Evidence custodian	Forensic examiner	Imaging	Seals opened, new seal no. noted	Both
5	D2 15:45	HD-01, USB-01, images	Forensic examiner	Evidence custodian	Return to storage; images stored	Originals resealed, new seal numbers	Both
6	D3 10:00	Working copy of image	Evidence custodian	Examiner	Laboratory examination	Hash verified before use	Both

Each time a seal is broken or re-applied, record the seal number, reason and who was present.

References

1. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
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5. Carrier, B. (2005). *File System Forensic Analysis*. Addison-Wesley.
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7. ILookIX / IXImager documentation (Perlustro LP): cite the version in your lab.

2.4 RECOVERY OF DELETED EVIDENCE FROM VOLUME SHADOW COPIES

Selected scenario: Recovering deleted internship-report evidence and browser traces from Volume Shadow Copies.

Assumptions

A1. A verified forensic image of a Windows 10 NTFS system is available. The suspect is an employee, referred to as **user E** (hypothetical).

A2. System Protection (restore points) was enabled, so Volume Shadow Copies (VSCs) exist on the system volume.

A3. The target is a document (*Internship_Report_Final.docx*, hypothetical name) and browser history for a specific site.

A4. All dates, times and file names are **hypothetical**. No real recovery results are claimed.

1. Selecting and categorizing evidence before detailed analysis

An image can hold hundreds of thousands of files. Investigators narrow it to what is relevant without destroying anything, since the full image is preserved.

Selection procedure

1. **Re-read the scope** from the authorization: user, device, time window, type of activity.
2. **Verify the image hash** and record tool versions.
3. **Exclude known files:** apply hash sets of standard operating-system and application files to filter noise.
4. **Filter by user, path and date range** (e.g., user E's profile, the period around the suspected activity).
5. **Filter by type and content:** documents, archives, browser data, keyword and file-name searches.
6. **Check for disguised files** using file-signature analysis.
7. **Include deleted and recovered items** (Recycle Bin, unallocated space, shadow copies).
8. **Tag and categorize** each item, record why it was included or excluded, and keep the filtering criteria in the notes.

Excluded data is *filtered from view*, not deleted. The reasoning is recorded so a reviewer can check it.

2. Evidence classification scheme

Table 1. Evidence classification table

Category	Definition	Examples in this case	Evidential weight / handling
User-created files	Content created or edited by the user	Report documents, spreadsheets, notes, saved images	High: examine content and metadata; compare versions
System/application files	Files generated by the OS or applications	Windows event logs, registry hives, Prefetch, application settings, Recycle Bin index files	Medium to high: show activity context (what ran, who was logged in, when)
Potentially suspicious files	Items whose type, location or behaviour is unusual	Renamed or mismatched extensions, encrypted archives, wiping/cleaning utilities, unknown executables	Needs investigation: do not assume intent; validate
Deleted/recovered artifacts	Items no longer visible in the normal file system but recovered	Files in Recycle Bin, carved files, older versions in shadow copies, deleted browser entries	High if validated; always record source (snapshot ID, offset)
Supporting/contextual evidence	Background that explains or corroborates	Installed software list, user accounts, USB device history, logon events, network/cloud sync settings	Supporting: strengthens other findings but rarely proves them alone
Excluded/irrelevant	Known system files or clearly out-of-scope data	Standard Windows files matched by hash set	Not analysed further; exclusion logged

3. Event Analysis for reconstructing activity

Event Analysis brings time-stamped records from many artifacts into one combined, sorted view so the sequence of actions becomes visible. The exact feature set depends on your ILookIX version; the general method is:

1. Collect events: file creation/modification/deletion times, Recycle Bin entries, registry key times, event-log entries, browser visits and downloads, USB connection times, program execution records.
2. Normalise all times to one reference time zone.
3. Sort by time and filter around the period of interest.
4. Group related events into sequences (e.g., browse, download, create file, copy to USB, delete).
5. Look for clusters, gaps (possible log clearing), and contradictions.
6. Validate key events using at least two independent artifacts.

Example use here: Event Analysis can show that a document was last modified, then a USB device connected, then the document deleted, then the browser history cleared, all within minutes. This sequence suggests an *activity pattern* but does not, by itself, prove who did it or why.

4. Cloud Analysis and Lead Analysis

Table 2. Purpose and questions answered

Technique	Purpose	Questions it may help answer
Cloud Analysis	Identify artifacts showing use of cloud storage, email or online services on the system, such as sync-client folders, account identifiers, cloud URLs in browser history, and cloud-related application files	Was the document uploaded or synced to a cloud account? Which cloud services and accounts did the user use? When did syncing occur? Is more evidence held outside the device and needs a legal request or separate authorization?
Lead Analysis	Extract and rank pieces of information (e.g., names, email addresses, URLs, phone numbers, keywords, user names) that point to further lines of enquiry and relationships between items	Who did the user communicate with? Which accounts, sites or contacts appear repeatedly? Which files or keywords connect to the incident? What should the investigator examine next?

Cloud Analysis helps find *where else* evidence might exist. Lead Analysis helps decide *what to examine next*. Both give leads that must be confirmed with direct evidence. Accessing cloud accounts requires proper legal authority.

5. How Volume Shadow Copy analysis helps

Volume Shadow Copy Service (VSS) is a Windows feature that creates point-in-time snapshots of an NTFS volume, using block-level storage of changes in the System Volume Information area. Snapshots are created, for example, by restore points, software installation or scheduled backups.

Value to the investigation

- **Recovery:** a file deleted or overwritten after a snapshot may still exist in its older state inside that snapshot.
- **Comparison:** the same file or registry/browser database can be compared across snapshots and the current volume to show what changed (e.g., content edited, history entries removed).
- **Time anchoring:** each snapshot has a creation time, showing the system's state at that moment.
- **Anti-forensic detection:** the absence of expected snapshots, or signs that snapshots were deleted, is itself a finding.

Limits: snapshots exist only if they were created and not deleted or removed automatically when storage limits were reached. Snapshots reflect state at one moment, so activity between snapshots may be missed.

Figure 1. Volume Shadow Copy recovery workflow

VERIFY: Verify the forensic image hash and record tool versions

↓

DETECT: Determine whether Volume Shadow Copies are present on the relevant volume

↓

INVENTORY: Record each snapshot identifier, creation time, and available metadata

↓

EXAMINE READ-ONLY: Open snapshots using a supported forensic tool or documented workflow

↓

COMPARE: Compare snapshots with the current volume and each other; identify missing/changed files and earlier browser data, registry hives, or logs

↓

EXTRACT AND HASH: Extract relevant items with metadata and calculate hashes for exported artifacts

↓

VALIDATE: Check provenance, metadata, content, consistency, and alternative explanations

↓

CORRELATE AND REPORT: Record snapshot ID and path; correlate with the timeline and document limitations

snapshot)

6. Hypothetical timeline: create, modify, copy, delete, recover

Table 3. Hypothetical event timeline (illustration only; times are invented)

Seq.	Time (IST)	Event	Evidence source	Notes
1	Day 1, 09:12	*Internship_Report_Final.docx* created in user E's Documents folder	File-system creation time	Creation time may reflect a copy
2	Day 1, 09:40	File modified and saved	Modification time; document metadata	Content version V1
3	Day 1, 10:05	File copied to a USB device	USB connection record; shortcut/recent-file entry	USB serial recorded
4	Day 2, 18:30	Restore point created (Volume Shadow Copy #2)	Snapshot creation time	Snapshot holds V1
5	Day 3, 22:10	File deleted and moved to Recycle Bin	Recycle Bin index entry	Deletion time recorded
6	Day 3, 22:12	Recycle Bin emptied	Missing items, file-system journal	Original no longer visible
7	Day 3, 22:15	Browser history cleared	Reduced or empty current history database	Older history may remain in snapshots
8	Day 4, 11:00	Forensic image acquired and verified	Acquisition log	Evidence preserved
9	Day 5, 14:20	Snapshot #2 opened; file recovered (V1)	Recovery workflow, Section 5	Recovered version reflects state at Day 2, 18:30

Interpretation: the recovered file shows the document *existed in that state* at the snapshot time. It does not say who deleted it or why. If edits happened after the snapshot (between Day 2, 18:30 and Day 3, 22:10), the recovered version may differ from the final version.

7. Validating a recovered artifact before reporting

Table 4. Validation checklist

Check	Why
Image hash verified before recovery	Evidence integrity
Snapshot ID and creation time recorded	Source and time anchor of the artifact
Recovered item hashed; hash recorded	Later verification
Extraction repeated (or done with a second tool) with the same result	Rules out tool error
File opens correctly; header/signature matches the extension	Confirms it is a valid file, not corrupt data
Metadata compared (size, timestamps, author fields) with the live volume or other snapshots	Shows how the version differs
Corroborated with independent artifacts (Recycle Bin, shortcut/recent-file entries, journal, event logs)	One source alone is weak
Time zone and clock offsets checked	Avoids timeline errors
Alternative explanations considered (system clean-up, auto-deletion, restore, sync)	Prevents over-claiming
Tool names and versions documented	Reproducibility
Findings reviewed by a second examiner (peer review)	Quality control
Wording of the finding states only what the evidence supports	Defensible report

8. Sources of error and alternative explanations

Table 5. Possible errors and alternative explanations

Area	Issue	Effect	Mitigation
Timestamps	Wrong system clock or time zone	Events appear shifted or out of order	Record clock offsets; normalise to one time zone
Timestamps	Copy or move operations alter times	Creation time may not mean original creation	Compare with other artifacts and journals
Timestamps	Deliberate manipulation	False timeline	Cross-check with logs and snapshots
Snapshot timing	Snapshot reflects a moment, not the final state	Recovered version may differ from what existed at deletion	State the snapshot time clearly
Snapshot timing	Snapshots deleted, expired or limited by storage	Missing history may be innocent or deliberate	Examine settings, event logs and storage limits
Deletion	File removed by the system (clean-up tools, antivirus, updates), not the user	Wrong attribution of deletion	Check logs and tools installed
Deletion	Emptying the Recycle Bin or deletion by another account	Wrong person blamed	Correlate with logon events and profile
Browser artifacts	Cleared by routine settings (e.g., auto-clear on exit)	History absence is not proof of concealment	Check browser settings
Recovered data	Carved or partial files, overwritten blocks	Incomplete or incorrect content	Validate; report as partial
Tools	Parsing bugs or version differences	Wrong interpretation	Second tool, documented version

Area	Issue	Effect	Mitigation
Attribution	Shared account or remote access	Account is not a person	Seek corroborating evidence

Conclusion

Volume Shadow Copies can preserve earlier states of files and browser artifacts that were deleted or changed. A defensible finding needs careful selection and classification of evidence, a read-only recovery from identified snapshots, event-based reconstruction of the sequence, and validation by independent checks. Findings should describe what existed at a known snapshot time and clearly state timestamp and attribution limitations.

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3. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
4. Microsoft. *Volume Shadow Copy Service* (Microsoft Learn documentation).
5. libyal project. *libvshadow: Volume Shadow Snapshot (VSS) format documentation*.
6. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
7. ILookIX documentation (Perlustro LP): confirm the exact Event, Cloud and Lead Analysis features in your installed version.

2.5 MULTI-SOURCE INVESTIGATION OF A SUSPECTED CYBER INCIDENT

Selected scenario: Unauthorized access to the college online assignment-submission portal using a student account, with evidence spread across a lab PC, browser, email, messaging, an Android phone and cloud services.

Assumptions

A1. The student account is referred to as **account S**. The incident is an unauthorized login to the portal followed by export of a file of student marks.

A2. Written authorization exists for the lab PC, the phone and requests to the portal administrators. Access to third-party cloud, email and messaging providers needs separate lawful process.

A3. The PC runs Windows 10 with a Chromium-based browser; the phone is an Android device; email is webmail; messaging is a mobile chat app. All are generic.

A4. All times are IST and all artifacts, names and IP addresses are **hypothetical**. No real findings are claimed.

1. Potential evidence from Windows and other operating systems

Table 1. Operating-system evidence sources

System	Evidence	What it can show
Windows	Security/system event logs	Logon and logoff of accounts, remote sessions, service and policy changes
Windows	Registry hives (NTUSER.DAT, SYSTEM, SOFTWARE)	Programs run, recent files, USB device history, installed software, time zone, Wi-Fi networks
Windows	Prefetch, shortcut (LNK) files, Jump Lists	Program execution and files opened by each user
Windows	Recycle Bin, file-system metadata, journal	Deleted files, creation/modification times
Windows	Page file, hibernation file, RAM capture	Remnants of passwords, URLs, chat text, running processes
Windows	Scheduled tasks, startup entries, installed applications	Persistence or tools used
Linux (e.g., portal server or a student's second device)	Authentication and web-server logs, shell history, cron jobs, home directory	Logins, commands run, files accessed
macOS (if a student device)	Unified logs, property lists, file-system event records, keychain	Application use, logins, file activity
Server OS / web application	Web-server access logs, application logs, session records	Source IP, user agent, URL and time of each request made as account S
Mobile OS (covered in Section 4)	App databases, system logs, Wi-Fi records	User activity and location of the phone

2. Browser artifacts and why they must be interpreted carefully

Table 2. Browser artifacts

Artifact	What it may reveal
History (URLs, titles, visit times, visit counts)	Sites and pages visited, including the portal login page
Downloads list	Files downloaded, source URL, save location, time
Cache and temporary files	Copies of pages, images and sometimes file content
Cookies and session data	Logged-in sessions, site preferences, account identifiers
Form data and saved logins	User names typed, saved credentials (stored encrypted)
Search terms	Queries entered into search engines
Bookmarks, extensions, session-restore files	Interests, tools in use, tabs open at last close
Synced data	History and passwords synchronised from other devices

Why interpretation needs care

- A visit record shows a page was loaded, not that the user read it or intended it (redirects, pop-ups, background tabs, link pre-loading).
- A shared computer or shared browser profile makes it hard to say who was browsing.
- Synchronisation can make history from another device appear on this one.
- Browsers store times in their own formats (e.g., microseconds since 1 January 1601 in Chromium-based browsers). Wrong conversion shifts the timeline.
- Private browsing, history clearing and automatic clean-up reduce evidence; absence of history is not proof of hiding.
- Cached or saved data may predate the incident.

3. Email and messaging evidence and correlation

Table 3. Email and messaging evidence and its correlation value

Evidence type	Details	How to correlate with other artifacts
Email headers	Sender, recipient, date, message ID, relay (Received) lines, sometimes the originating IP	Compare originating IP and time with portal logs and campus DHCP/Wi-Fi logs
Email body and attachments	Content, file names, embedded links	Hash an attachment and compare with the file found on the PC or in the browser downloads
Mailbox and cache files (e.g., local mail stores, webmail cache)	Sent, deleted and draft items	Match sent times with browser sessions and logon events
Calendar and contacts	Meetings, contacts, relationships	Show associations and context
Chat database on phone/PC	Messages, timestamps, sender/receiver IDs, group names	Compare message times and mentions of the file or login with the timeline
Shared media and attachments	Images, documents sent through chat	Hash and compare with PC and cloud files
Call logs and voice/video metadata	Who communicated with whom and when	Show coordination between persons
Provider-side records	Login history, IP addresses, message metadata (through lawful process)	Confirm device and location of access

Headers can be forged, and sender names are easily spoofed, so headers should be validated against server-side records where available.

4. Forensic considerations for the mobile phone

Table 4. Mobile examination considerations

Consideration	Action / reason
Legal authority and scope	Phones hold large amounts of personal data, so examine only what the authorization covers
Isolation	Prevent remote wiping and new data: airplane mode if the device is on and unlocked, or a Faraday bag. Record the power and lock state
Do not power cycle unnecessarily	Powering off may lock the device and lose volatile data and keys
Acquisition level	Logical, file-system, physical or chip-level methods are possible, depending on device, operating system version and lock status. State which was used and why
Encryption and passcode	Modern phones use strong encryption. Without the passcode, extraction may be limited
Hashing and documentation	Record device identifiers (e.g., IMEI), tool and version, acquisition type and hash of the extraction
Cloud and backups	Phone data may sync to cloud accounts, which need separate authority
Time handling	Record the phone's clock and time zone against a reference time

Challenges with mobile evidence

- Wide variety of devices, operating system versions and applications, with frequent updates.
- Strong encryption and locked devices.
- Application data in protected locations, in changing formats; tool support may lag behind new app versions.
- Disappearing or deleted messages and remote-wipe risk.
- Data mixes personal and case-related content, so privacy handling is needed.
- Evidence may also live in backups and cloud storage, outside the device.

5. Internet and cloud-based evidence sources

Table 5. Internet and cloud sources

Source	Evidence	How obtained / limits
Portal web-server and application logs	Login attempts, source IP, user agent, accessed pages and file exports for account S	Institution's own administrators; preserve and hash copies early
Authentication/single sign-on and multi-factor logs	Successful and failed logins, device and location data	Institution or identity provider
Campus network logs (DHCP, Wi-Fi/RADIUS, firewall, proxy, DNS)	Maps IP address to device (MAC) and time; shows connections	Institution; retention periods vary
Internet service provider records	Subscriber details for an IP address	Lawful request through proper authority
Email provider records	Account login history, IP addresses, message metadata	Lawful process; provider retention varies

Source	Evidence	How obtained / limits
Cloud storage and productivity services	Upload and sharing logs, file versions, deleted items	Account owner's export or lawful process
Messaging and social media providers	Account records, message metadata	Lawful process; end-to-end encryption limits content
Third-party and archive sources	Public posts, cached pages, archived copies	Open-source collection; document method and time of capture

Cloud evidence should be preserved early, because retention periods are limited and users can delete data. Data may sit in other jurisdictions, which complicates access.

Table 6. Cross-source correlation matrix (artifacts, times in IST)

ID	Source	Hypothetical artifact	Time	Links to	What the link shows	Caution
C1	Internet/cloud (portal log)	Successful login as account S from IP *a.b.c.d*; later request to export a marks file	21:03; 21:15	C2, C3, C4	The account was used from this IP	Account is not a person
C2	Internet (campus DHCP/Wi-Fi log)	IP *a.b.c.d* leased to MAC of the lab PC	20:58	C1, C3	Ties the IP to a specific computer	Check lease times and clock offset
C3	Computer (Windows event log)	Logon by a local user account on the lab PC	20:56	C2, C4	Someone was logged on locally at that time	Shared lab account possible
C4	Browser (history and downloads)	Portal login page visited, then *marks_export.csv* downloaded	21:03; 21:15	C1, C5	Browser on this PC fetched the file	Visit is not proof of who typed
C5	Email (webmail sent item)	Message from S's mailbox with attachment of the same file name	21:30	C4, C6	File left the PC by email	Header time vs. server time
C6	Computer + email	Hash of the attachment equals the hash of the downloaded file	n/a	C4, C5	Same file content in both places	A hash match proves content identity only
C7	Mobile (chat database)	Message to a contact referring to "the file"	21:35	C5	Possible sharing or coordination	Message wording is ambiguous
C8	Mobile (Wi-Fi connection record)	Phone connected to the same campus Wi-Fi	20:45	C2, C7	Phone and PC were on the same network	Presence on a network is not use
C9	Cloud (storage activity log)	Upload of the same file from IP *a.b.c.d*	21:40	C1, C6	Possible onward copy	Needs provider confirmation

6. Major challenges in multi-device and cloud forensics

Table 7. Challenge analysis

#	Challenge	Why it matters	Mitigation
1	Data volume and variety	Many devices and formats make review slow and error-prone	Use triage, filtering and timelines; document criteria
2	Encryption and locked devices	Content may be unreadable	Seek lawful key/passcode access; rely on metadata and server-side records
3	Cloud data location and jurisdiction	Data may be held abroad or by third parties	Early preservation requests; lawful process; legal advice
4	Limited control and retention	Providers keep logs for limited periods; users can delete	Send preservation requests quickly
5	Time synchronisation	Each source has its own clock, time zone and format	Normalise to one reference; record offsets
6	Chain of custody for remote data	No physical item to seal; data may change	Record method, time, account and hash of exports

#	Challenge	Why it matters	Mitigation
7	Authenticity and integrity of provider-supplied data	Hard to verify what the provider returns	Document source and request method; cross-check with other sources
8	Shared accounts and multiple users	Account does not equal person	Corroborate with physical presence, devices and logs
9	Privacy and legal constraints	Devices and cloud accounts hold personal data	Limit scope to authorization; follow data-protection law
10	Anti-forensic actions	Deletion, wiping, VPN use, private browsing	Look for traces in backups, logs and other devices
11	Tool and version compatibility	Apps and formats change fast	Use validated tools; test with a second tool

7. Final investigation conclusion (based on the hypothetical artifacts above)

This conclusion illustrates the structure of a final summary. Real conclusions depend on real evidence.

Table 8. Final investigative summary

Category	Content
Direct evidence	Portal log (C1) records account S logging in and exporting the marks file. Webmail sent item (C5) records a message with that file attached from S's mailbox. Both show activity by the *account* directly
Corroborating evidence	DHCP/Wi-Fi log ties the IP to the lab PC (C2). Windows logon (C3) and browser history and download (C4) place the activity on that PC. Hash match (C6) shows the exported and emailed files are identical. Phone Wi-Fi record (C8) and chat message (C7) are consistent with the same period. Cloud upload (C9) suggests an onward copy
Unresolved questions	Which person was at the keyboard (shared lab account or shared credentials). How the credentials were obtained, and whether the account was compromised. Whether the phone user and the PC user are the same person. Whether deleted messages or files exist. Provider-side confirmation of the cloud upload. Clock offsets between sources

Conclusion :

The combined artifacts support, with moderate-to-high confidence, that account S was used from the lab PC to access the portal, export a marks file and send it onward. They do not, by themselves, identify the individual operating the PC. Further steps: obtain provider records through lawful process, review lab access records, and interview relevant persons under proper procedure.

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